

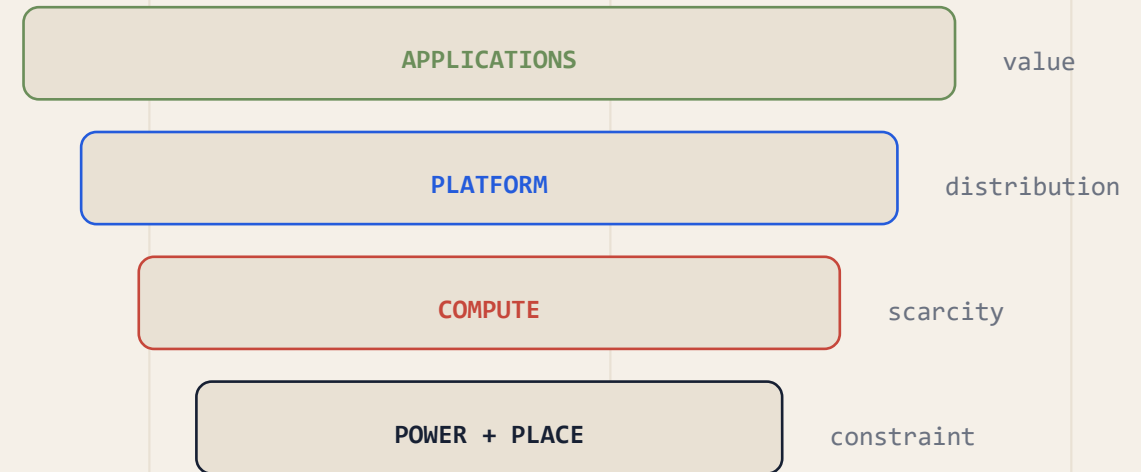
# AI INFRASTRUCTURE ECONOMICS

Capital moves first. Constraints decide what ships.

**A report on capacity, capital, and the physical stack.**

A source-led editorial data study  
for strategy and platform operators.

THE VALUE CHAIN IS PHYSICAL



# AI is not one market. It is a stack of dependencies.

The margin pool can move upward; the bottleneck often stays physical.

## 07 / APPLICATIONS

workflow · distribution · willingness to pay

## 06 / MODELS

training · inference · evaluation

## 05 / SOFTWARE

orchestration · serving · observability

## 04 / SYSTEMS

accelerator · memory · network fabric

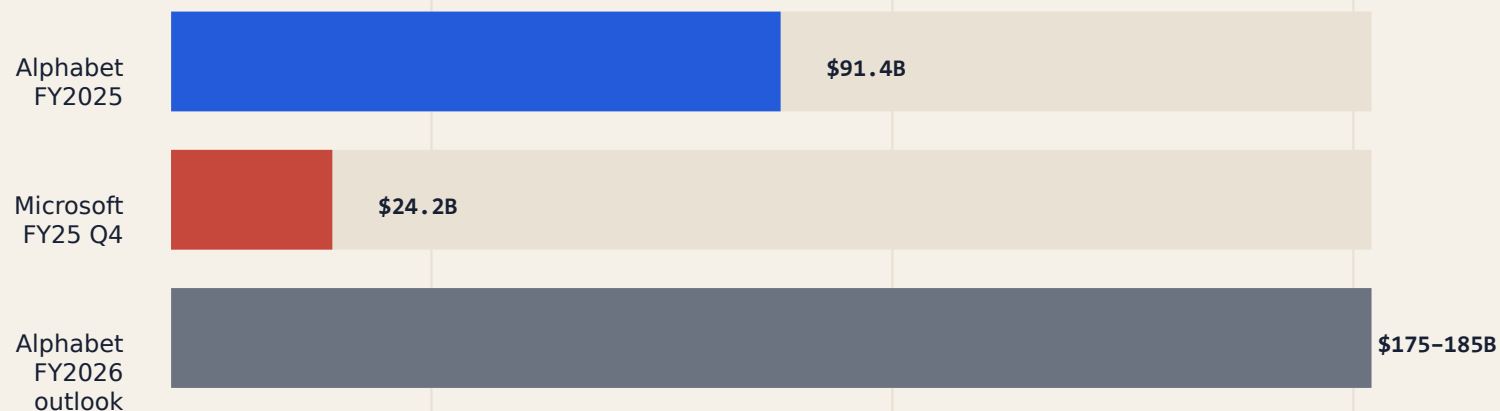
## 03 / DATA CENTERS

land · cooling · power delivery

dependency ↑

# Capital is accelerating — but the periods are not interchangeable.

Reported figures are shown with company and fiscal-period labels, not as a false league table.



## READ THIS AS

directional evidence  
not a single-period ranking  
different disclosure periods  
different business mix

### WHAT IS REPORTED

Large technical-infrastructure commitments are visible.

### WHAT IT SUGGESTS

Capacity is becoming a strategic balance-sheet item.

### WHAT IT DOES NOT PROVE

CapEx alone does not tell us who captures the margin.

# Where the money lands changes the depreciation story.

One disclosed split makes the physical stack visible.

ALPHABET FY2025

## \$91.4B

technical infrastructure CapEx

**The stack has  
different clocks.**

A server cycle can move faster  
than a building or grid connection.

APPROXIMATE 2025 COMPOSITION

60% SERVERS

40% DC + NETWORK

### SHORTER CYCLE

servers  
faster refresh  
utilization pressure

### LONGER DURATION

data centers  
networking  
depreciation horizon

# The constraint is not “compute.” It is the coordination of four physical systems.

Land, energy, networking, and servers arrive on different timelines.

## ARRIVAL ORDER

01

02

03

04

### LAND

permitted + buildable  
lead time  $\neq$  demand

### ENERGY

predictable supply  
lead time  $\neq$  demand

### NETWORK

fabric + interconnect  
lead time  $\neq$  demand

### SERVERS

GPU + components  
lead time  $\neq$  demand

**CAPACITY ONLY SHIPS WHEN THE SLOWEST LAYER ARRIVES.**

This is why supply-chain risk belongs inside the AI strategy — not in a footnote after the model roadmap.

# Capacity becomes economics only through utilization.

The flywheel is simple to draw — and difficult to keep turning.

## CAPACITY

hardware + power

## UTILIZATION

load + scheduling

## REVENUE

serving demand

## CASH

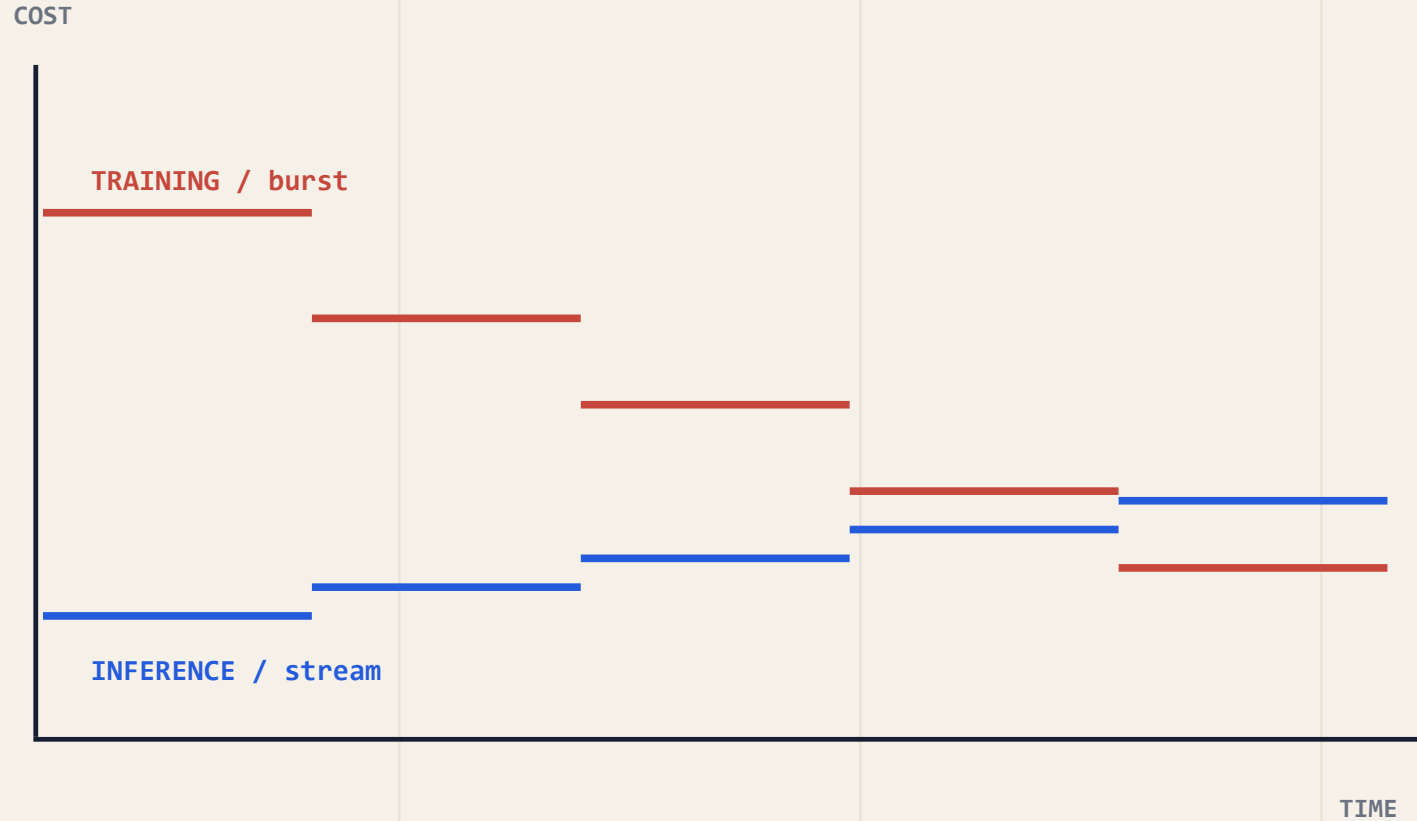
depreciation + margin

OPERATING  
LEVERAGE

NO UTILIZATION → NO LEVERAGE

# Training is episodic. Inference is continuous.

Illustrative curves show why utilization and serving efficiency can matter more than a single benchmark.



## OPERATOR QUESTION

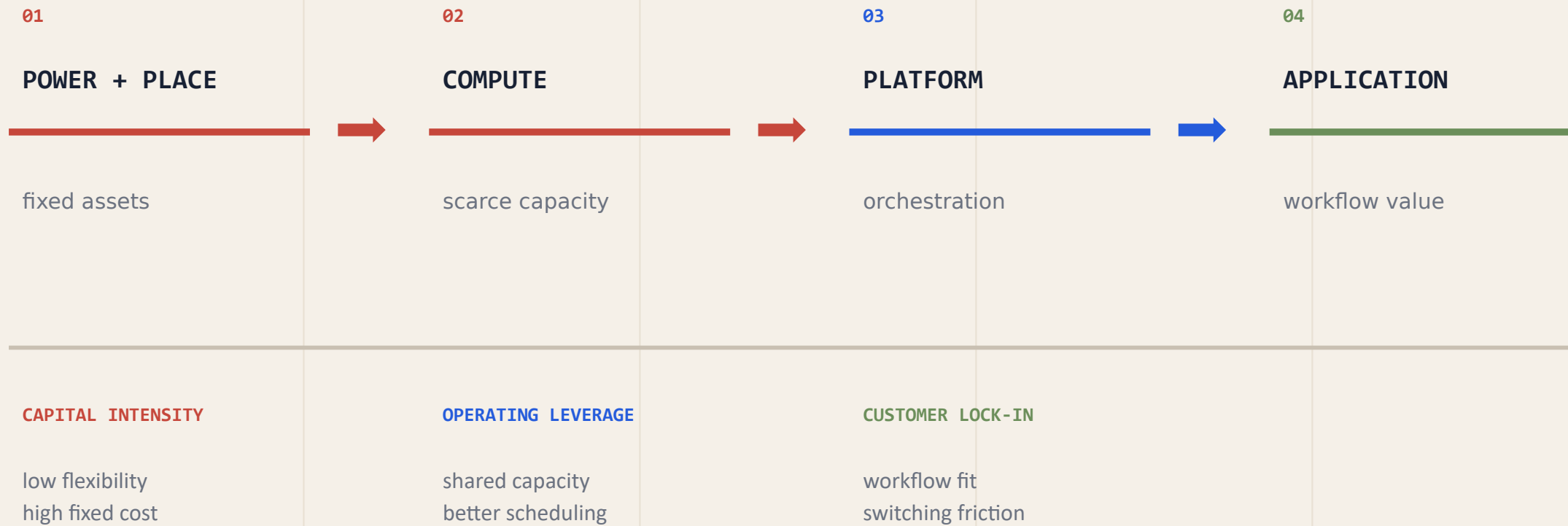
Can the load be scheduled, shifted, or pooled?

## ILLUSTRATIVE ONLY

Shape of curve is conceptual not a benchmark

# The margin pool is a flow, not a single layer.

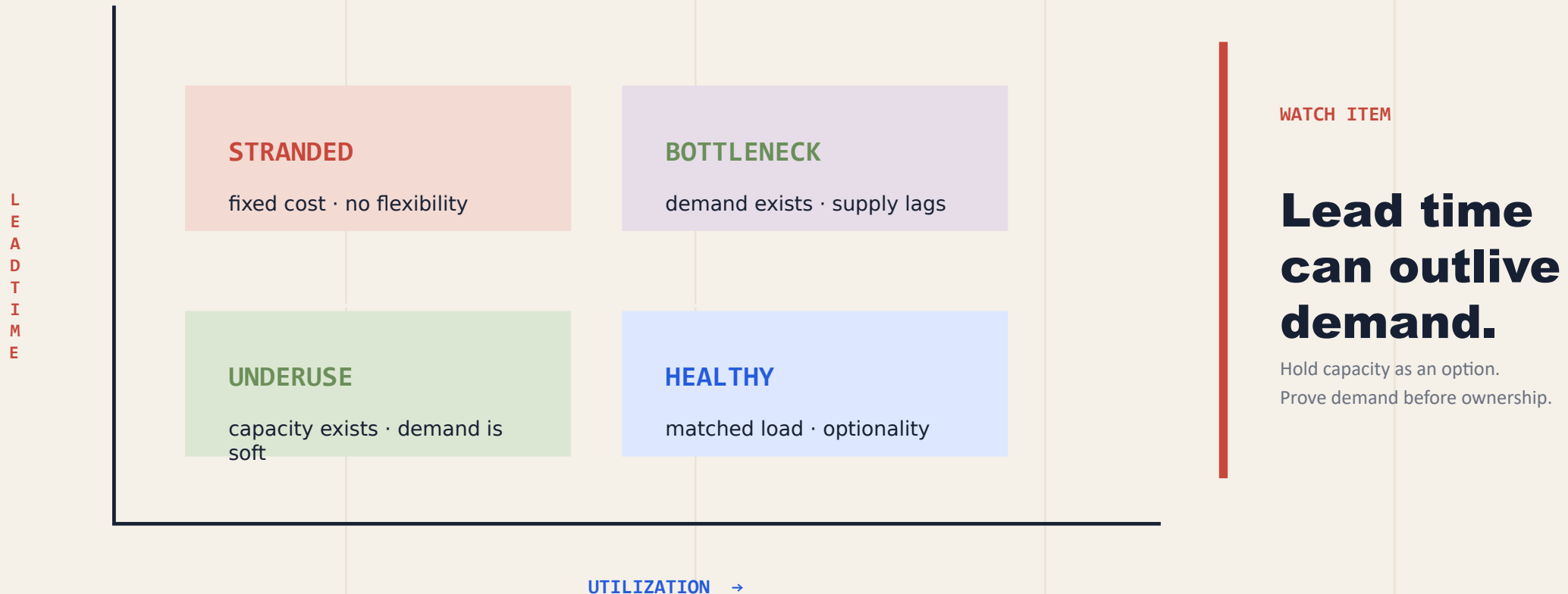
Compute creates the possibility; platforms and applications decide where value compounds.





# The dangerous asset is capacity you cannot use.

A simple matrix makes the hidden risk visible: high lead time plus low utilization.



# There are three ways to buy capacity. None is free.

The right answer depends on demand certainty, capital appetite, and control requirements.

| BUY     |        | RENT        |  | BUILD           |  |
|---------|--------|-------------|--|-----------------|--|
| control |        | optionality |  | differentiation |  |
| CAPITAL | high   | variable    |  | very high       |  |
| SPEED   | medium | fast        |  | slow            |  |
| CONTROL | high   | shared      |  | highest         |  |
| RISK    | demand | vendor      |  | execution       |  |

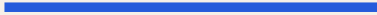
OPERATOR TEST

How certain is demand?  
How unique is the workload?

# Run infrastructure like an operating system, not a warehouse.

Five measures reveal whether the asset is earning its keep.

01



## CAPACITY

what exists

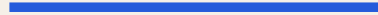
02



## UTILIZATION

what is used

03



## LATENCY

what users feel

04



## DEPRECIATION

what time costs

05



## POWER

what scale requires

## THE SCORECARD TURNS CAPEX INTO A FEEDBACK SYSTEM.

If one measure is missing, the operator is managing a blind spot — not a platform.

# The winning infrastructure is not the biggest. It is the most legible.

Capital creates the option. Constraints define the clock. Control decides the return.

## Capital creates the option.

## Constraints define the clock.

## Control decides the return.

### THE OPERATOR'S QUESTION

Where is the bottleneck?

What can we flex?

What must we own?

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MAKE THE STACK LEGIBLE.